Storage Devices

What is a Storage Device?

A storage device is hardware that **permanently stores data, programs and the operating system**, even when the computer is switched off. It is also called **secondary memory** or **auxiliary storage**. The CPU cannot access it directly, so data is first copied into RAM and then processed.

Why we need it: RAM is temporary and loses data on power off, so we need storage to keep documents, photos, software and the OS.

Classification of Storage Devices

1. **Magnetic storage:** HDD, magnetic tape, floppy disk
2. **Optical storage:** CD, DVD, Blu-ray
3. **Solid-state (flash) storage:** SSD, pen drive, memory card
4. **Cloud storage:** data kept on remote servers

1. Magnetic Storage

Data is stored as magnetized patterns (north and south poles representing 0 and 1) on a coated surface.

Hard Disk Drive (HDD)

Parts of an HDD

- **Platters:** circular aluminum or glass disks coated with magnetic material
- **Spindle and motor:** spins the platters (5400, 7200 or 10000 RPM)
- **Read/write heads:** float just above the platter surface to read and write data
- **Actuator arm:** moves the heads across the platter
- **Controller board (PCB):** controls all operations and communicates with the computer

How data is organized

- **Track:** a circular ring on the platter
- **Sector:** a small section of a track (usually 512 bytes or 4 KB)
- **Cylinder:** the same track on all platters stacked together
- **Cluster:** a group of sectors, the smallest unit the OS allocates

Disk capacity = Heads × Tracks × Sectors per track × Bytes per sector

Performance terms

- **Seek time:** time for the head to reach the correct track

- **Rotational latency:** time for the sector to rotate under the head
- **Access time** = Seek time + Rotational latency
- **Data transfer rate:** how fast data moves (MB/s)

Advantages: cheap per GB, very large capacity (up to many TB)
Disadvantages: slower, noisy, uses more power, easily damaged by shock, and has moving parts that wear out

Form factors: 3.5-inch (desktops), 2.5-inch (laptops)

Magnetic Tape

- Long plastic ribbon with magnetic coating, read sequentially
- Cheap and durable for **backups and archives** (banks, data centers)
- Very slow, because data must be read in order (sequential access)

Floppy Disk (obsolete)

- Small portable disk, 1.44 MB capacity, no longer used

2. Optical Storage

Data is written by a **laser** that burns tiny pits (holes) and lands (flat areas) onto a reflective disc. A laser reads them back by detecting reflection differences.

Type	Capacity	Laser	Use
CD (Compact Disc)	650-700 MB	Infrared	Music, small data
DVD (Digital Versatile Disc)	4.7 GB (single layer), 8.5 GB (dual layer)	Red	Movies, software
Blu-ray (BD)	25 GB (single layer), 50 GB (dual layer)	Blue-violet	HD movies, large backups

Disc varieties

- **ROM:** read-only (factory pressed)
- **R:** recordable once
- **RW:** rewritable many times

Advantages: cheap, portable, good for distribution
Disadvantages: low capacity, slow, scratches damage them, becoming obsolete

3. Solid-State Storage (Flash Storage)

Uses **NAND flash memory chips** with no moving parts. Data is stored by trapping electrical charge in memory cells.

SSD (Solid State Drive)

Parts: NAND flash chips, a controller chip, and a DRAM cache.

Types of SSD by interface

Type	Interface	Speed (approx.)
SATA SSD	SATA III, 2.5-inch	About 550 MB/s
M.2 SATA	SATA, small card	About 550 MB/s
NVMe SSD (M.2)	PCIe	3,000 to over 12,000 MB/s depending on generation

NAND cell types (how many bits per cell)

- **SLC:** 1 bit, fastest and most durable, expensive
- **MLC:** 2 bits
- **TLC:** 3 bits, most common
- **QLC:** 4 bits, cheapest and highest capacity but slower and less durable

Advantages: very fast, silent, low power, shock resistant, quick boot times
Disadvantages: more expensive per GB, limited write cycles over its lifetime

Pen Drive (USB Flash Drive)

- Portable flash storage plugged into a USB port
- Capacity from 4 GB to 1 TB or more
- Speed depends on the USB version (USB 2.0, 3.0, 3.1, 3.2)

Memory Cards

- **SD card, microSD card:** used in cameras, phones and drones
- Classes and speed ratings (Class 10, UHS-I, UHS-II) show write speed

eMMC

- Soldered flash storage in budget laptops and phones, slower than SSD

4. Cloud Storage

Data is stored on internet servers instead of your device, and can be accessed from anywhere.

- **Examples:** Google Drive, OneDrive, Dropbox, iCloud
- **Advantages:** access anywhere, automatic backup, easy sharing
- **Disadvantages:** needs internet, subscription cost for large space, privacy concerns

Other Storage Concepts

RAID (Redundant Array of Independent Disks): combining multiple disks for speed or safety.

Level	Description
RAID 0	Striping: data split across disks for speed, no safety
RAID 1	Mirroring: identical copy on two disks for safety
RAID 5	Striping with parity: speed plus protection against one disk failure
RAID 10	Mirroring plus striping: both speed and safety

Other terms

- **NAS (Network Attached Storage):** a storage device shared over a network
- **SAN (Storage Area Network):** high-speed network of storage devices for data centers
- **External drives:** portable HDD/SSD connected through USB or Thunderbolt
- **Partitioning:** dividing a disk into logical sections (C:, D:)
- **Formatting:** preparing a disk with a file system (NTFS, FAT32, exFAT, ext4)

HDD vs SSD vs Optical

Feature	HDD	SSD	Optical
Technology	Magnetic	Flash memory	Laser
Speed	Slow to medium	Very fast	Slow
Moving parts	Yes	No	Yes
Durability	Fragile	Strong	Scratch-prone
Capacity	Very high	High	Low
Cost per GB	Low	Higher	Very low
Best for	Bulk storage, backups	OS, apps, speed	Distribution, archives

Power Supply

What is a Power Supply?

A power supply is the hardware that **converts electricity from the wall outlet into the correct voltages needed by computer components** and delivers it safely and steadily. Without it, no part of the computer can work.

Types of Power Supply in Computing

1. SMPS / PSU (Internal Power Supply Unit)

I explained the working of the SMPS earlier, so here is a quick recap with the extra details.

- Converts **AC (230V in India, 50 Hz)** to **DC (+3.3V, +5V, +12V)**
- Located at the top or bottom rear of the cabinet
- Cooled by a built-in fan

Form factors

- **ATX:** standard desktop size
- **SFX:** small form factor for compact PCs
- **TFX / Flex ATX:** for slim cases
- **Redundant PSU:** two PSUs in servers so one takes over if the other fails

Types by cable design

Type	Description
Non-modular	All cables permanently attached; cheapest, but messy
Semi-modular	Essential cables attached, others detachable
Fully modular	All cables detachable; best cable management

Safety protections in a good PSU

- **OVP:** Over Voltage Protection
- **UVP:** Under Voltage Protection
- **OCP:** Over Current Protection
- **OPP:** Over Power Protection
- **SCP:** Short Circuit Protection
- **OTP:** Over Temperature Protection

Choosing the right wattage: add up the power use of the CPU, GPU and other parts, then add 20-30% headroom. A basic office PC may need around 350-450W, while a powerful gaming PC may need 650-850W or more.

Efficiency (80 PLUS): certification levels are Standard, Bronze, Silver, Gold, Platinum and Titanium. For example, an 80 PLUS Bronze unit is at least 82-85% efficient at typical loads, meaning less electricity is wasted as heat.

PSU Connectors

Connector	Powers
24-pin ATX	Motherboard
4+4 pin EPS	CPU
6/8-pin PCIe	Graphics card
SATA power	HDD, SSD, optical drive
Molex	Old drives, fans
Floppy (Berg)	Old floppy drive

2. UPS (Uninterruptible Power Supply)

A UPS is a **backup power device** that supplies electricity for a short time when the main power fails, so you can save your work and shut down safely.

Parts: battery, inverter (DC to AC), charger and rectifier, and a switching circuit.

Types of UPS

Type	Working	Use
Offline / Standby	Runs from mains; switches to battery during a cut (small delay)	Home PCs
Line-interactive	Has a voltage regulator (AVR) that corrects small fluctuations; faster switching	Small offices
Online (double conversion)	Always runs from the battery/inverter; zero transfer time and cleanest power	Servers, hospitals, data centers

UPS rating: measured in **VA (Volt-Ampere)**, for example 600 VA, 1 kVA. Backup time depends on the load and battery capacity.

3. Voltage Stabilizer

Keeps the voltage steady by correcting **high and low voltage** from the mains. It does not provide backup during a power cut.

4. Surge Protector / Spike Guard

Protects devices from sudden voltage spikes (lightning, grid switching). It is a basic protection for the power strip.

5. Laptop Adapter and Battery

- **Adapter:** converts AC to DC (commonly 19V) to charge the laptop and run it
- **Battery:** usually Lithium-ion or Lithium-polymer; rated in Wh (watt-hours) or mAh
- Battery care: avoid keeping it at 100% or 0% for long, avoid heat

6. CMOS Battery

A small **CR2032 3V coin cell** on the motherboard that keeps the BIOS settings and system clock when the computer is off. If it dies, the date and time reset.

AC vs DC

AC (Alternating Current)	**DC (Direct Current)**
Direction changes periodically	Flows in one direction
Supplied by the wall outlet	Needed by computer components
Example: 230V, 50 Hz	Example: +12V, +5V, +3.3V

Common Power Problems and Protection

Problem	**Cause**	**Protection**
Power failure (blackout)	Total loss of power	UPS
Brownout (sag)	Low voltage	Stabilizer, UPS with AVR
Surge / spike	Sudden high voltage	Surge protector
Noise (EMI/RFI)	Electrical interference	Line conditioner, quality PSU
Overvoltage	High mains voltage	Stabilizer

Troubleshooting Power Supply Problems

- **PC doesn't start:** check the power cable, switch on the PSU, wall socket and the 24-pin connector
- **Random shutdowns or restarts:** possible overloaded or failing PSU, or overheating
- **Burning smell or noise:** switch off immediately and replace the PSU
- **Testing:** use a PSU tester or the **paper clip test** (an expert-only check)

Safety tips

- Never open the PSU; its capacitors can hold a dangerous charge
- Use a proper earthing (ground) connection
- Don’t overload extension boards
- Buy a PSU from a reputable brand with protection features